



- 1 Photosynthesis occurs in chloroplasts. A chloroplast suspension can be made by extracting the chloroplasts from green leaves.

The blue dye DCPIP can be used to investigate photosynthesis. During photosynthesis the blue colour of the DCPIP changes and eventually disappears so that it becomes colourless. The remaining mixture appears green due to the chloroplast suspension.

A student investigated the effect of light on photosynthesis in a chloroplast suspension.

- Step 1 The student crushed eight green leaves in an ice-cold buffer solution.
- Step 2 The mixture of crushed leaves and buffer solution was then filtered to form a chloroplast suspension.
- Step 3 The student was given two test-tubes. One test-tube was wrapped in aluminium foil to exclude light. The second test-tube was **not** covered in foil so that light could reach the contents of this test-tube.
- Step 4 A plastic pipette was used to add 2 cm^3 of the chloroplast suspension to each of the two test-tubes.
- Step 5 A syringe was used to add 5 cm^3 of DCPIP solution to both test-tubes. A stopper was placed in each test-tube.
- Step 6 Both test-tubes were placed in a water-bath containing ice-cold water. The water-bath was positioned near a bright light.
- Step 7 The test-tubes were left in the bright light for five minutes.
- Step 8 After five minutes the student observed and noted the colour of the liquid in the test-tube that was **not** wrapped in foil. They removed the foil from the covered test-tube and observed the colour in this test-tube.

The student's results are shown in Fig. 1.1.

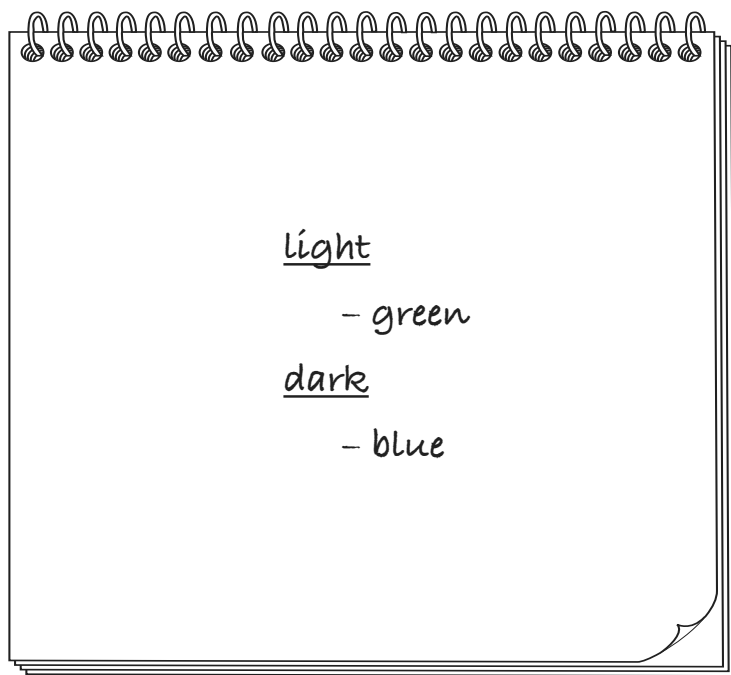


Fig. 1.1

- (a) (i) Prepare a table and record the results from Fig. 1.1.

[2]

- (ii) Identify the variable that was changed (independent variable) and the variable that was measured (dependent variable) in this investigation.

independent variable

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dependent variable

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[2]

- (iii) State **two** variables that were kept constant in this investigation.

1

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2

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[2]

- (iv) State a conclusion for these results.

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..... [1]

- (b) In step 6 both test-tubes were placed in a water-bath containing ice-cold water for five minutes. A thermometer was also placed in the water-bath.

Fig. 1.2 shows the thermometer at the start of the investigation and after five minutes.

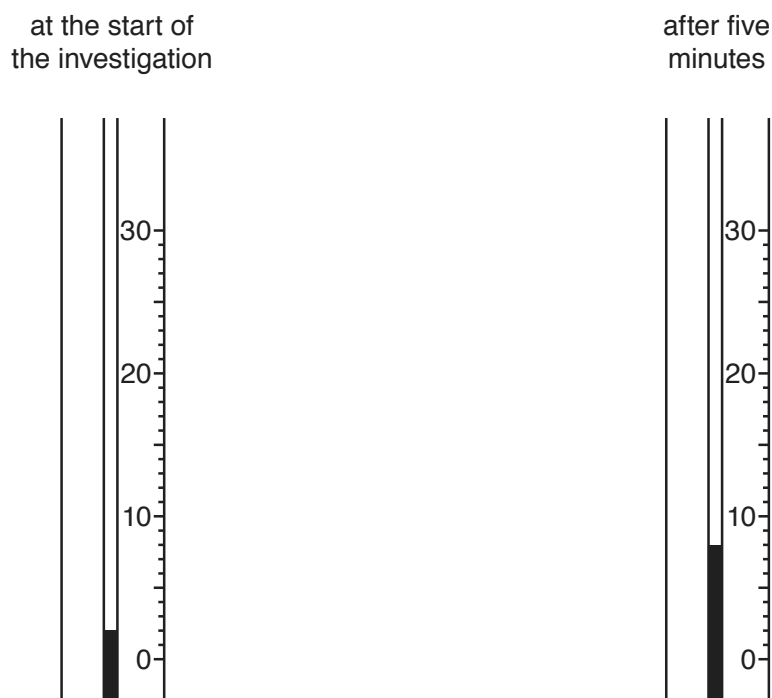


Fig. 1.2

Record the temperatures shown in Fig. 1.2. Include the units.

temperature at the start of the investigation

temperature after five minutes [2]

- (c) The investigation was not repeated.

- (i) Suggest why it would be advisable to repeat the investigation.

.....

 [1]

- (ii) Identify **one** other source of error in this investigation and suggest an improvement for this error.

error.....

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improvement

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[2]

- (d) A student wanted to investigate the effect of temperature on the rate of photosynthesis in a chloroplast suspension.

Describe how the student could carry out this investigation using DCPIP.

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..... [6]

[Total: 18]

- 2 Fig. 2.1 is a photomicrograph of part of the lower epidermis of a leaf.

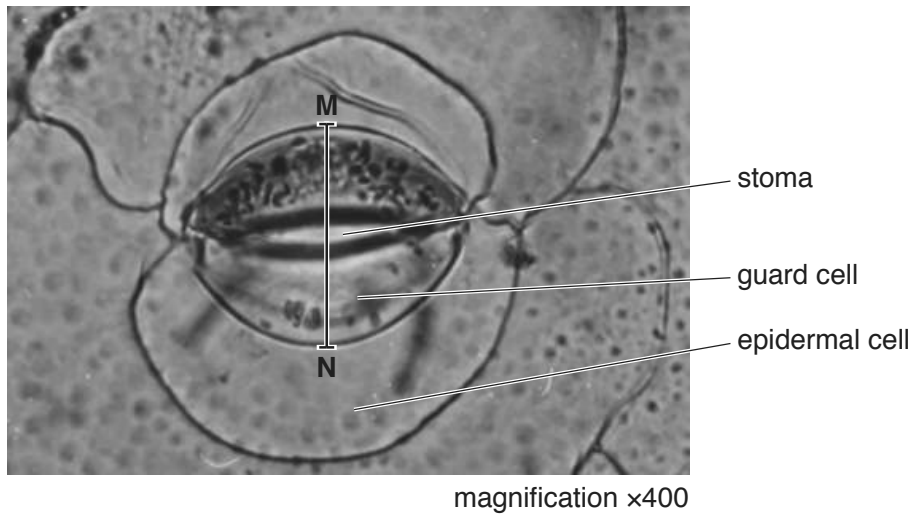


Fig. 2.1

- (a) (i) Make a large drawing of Fig. 2.1. Do not label your drawing.

[4]

- (ii) Measure the length of the line **MN** on Fig. 2.1.

length of line **MN** mm

Calculate the actual width of the guard cells and the stoma indicated by line **MN** in Fig. 2.1. Use the equation and information in Fig. 2.1.

$$\text{magnification} = \frac{\text{length of line MN on Fig. 2.1}}{\text{actual width of the guard cells and the stoma}}$$

Give your answer to two decimal places.

..... mm
[3]

- (b) A student investigated the effect of sucrose concentration on stomatal opening.

Strips of epidermis from leaves were placed in different concentrations of sucrose solution for one hour.

The student measured the width of 10 stomatal openings from the epidermis in each sucrose solution.

The results are shown in Table 2.1.

Table 2.1

percentage concentration of sucrose	width of stomatal opening / μm										average
	1	2	3	4	5	6	7	8	9	10	
0	5	6	7	5	5	6	7	5	7	6	5.9
5	3	4	3	3	4	3	2	3	3	2	
20	1	2	1	2	1	1	5	1	2	1	1.7

- (i) Circle **one** measurement in Table 2.1 that could be anomalous. [1]

- (ii) Calculate the average width of stomatal opening in the 5% sucrose solution.

Include the unit.

.....
[2]

- (c) Pineapple plants are adapted to grow in hot dry conditions.

Scientists investigated the number of stomata open, in pineapple plants, at different times of day.

The data is shown in Table 2.2 (00:00 is midnight).

Table 2.2

time of day	average number of stomata open per mm ²
00:00	77
04:00	61
08:00	22
12:00	4
16:00	10
20:00	51

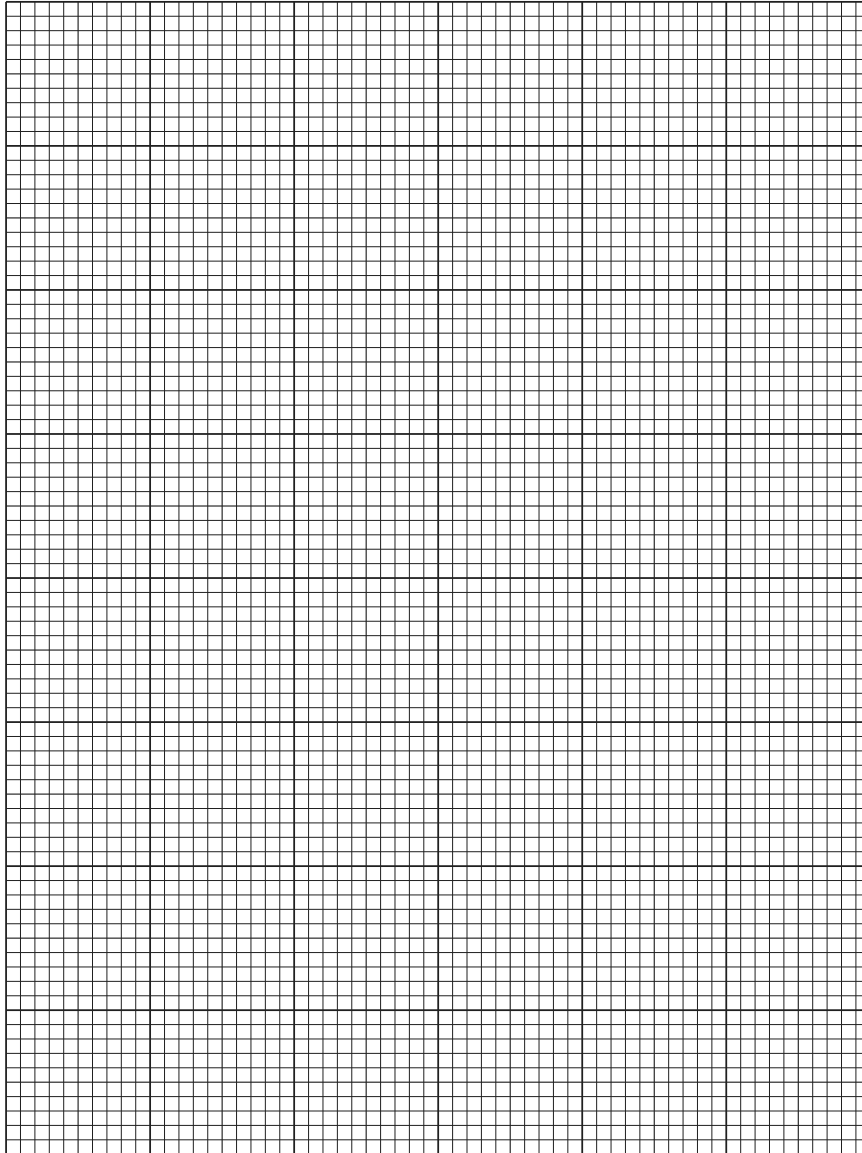
- (i) Calculate the percentage change in the average number of open stomata per mm² between 16:00 hours and 20:00 hours.

Space for working.

.....%

[2]

- (ii) Plot a bar chart on the grid to show the data in Table 2.2.



[3]

- (iii) Describe, using your graph, how the average number of open stomata per mm^2 changes throughout the day.

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..... [2]

(d) When fruit ripens the starch contained within it is converted into reducing sugars.

(i) State the name of the substance that is used to test for the presence of starch.

..... [1]

(ii) Describe how you could safely show a reducing sugar was present in a sample of pineapple fruit juice.

procedure

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safety.....

..... [4]

[Total: 22]

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